

Problem

The joint density for the random variable (X, Y) where X is the temperature change and Y is the proportion of the spectrum that shifts for a certain atomic particle is

$$f(x, y) = \begin{cases} 10xy^2 & 0 < x < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

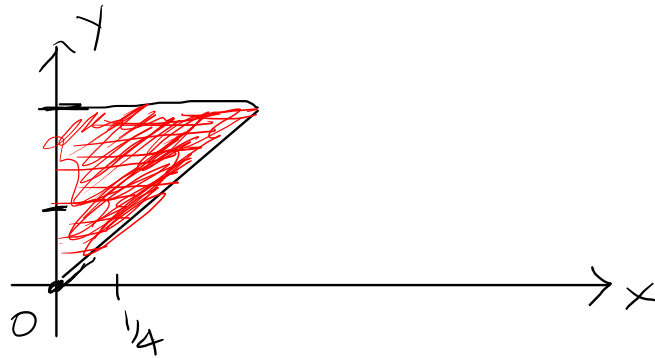
1) Find $f_x(x)$, $f_y(y)$ and $f(y|x)$

2) Find the probability that the spectrum shifts more than half of the total observations, given that the temperature is increased by 0.25 units.

Soln

$$f_x(x) = \int_{-\infty}^{\infty} f(x,y) dy = \int_x^1 10xy^2 dy = \left[\frac{10xy^3}{3} \right]_x^1$$

$$= \frac{10x}{3} - \frac{10x^4}{3} = \frac{10x(1-x^3)}{3}; \quad 0 < x < 1$$



$$f_y(y) = \int_{-\infty}^{\infty} f(x,y) dx = \int_0^y 10xy^2 dx = 5y^4; \quad 0 < y < 1$$

$$f(y|x) = \frac{f(x,y)}{f_x(x)} = \frac{10xy^2}{\frac{10x(1-x^3)}{3}} = \frac{3y^2}{1-x^3}, \quad 0 < x < y < 1$$

$$f(x|y) = \frac{f(x,y)}{f_y(y)} = \frac{10xy^2}{5y^4} = \frac{2x}{y^2}, \quad 0 < x < y < 1$$

$$P(Y > \frac{1}{2} | X = 0.25) = \int_{\frac{1}{2}}^1 f(y | x = 0.25) dy$$

$$\left[P(a < Y < b | X = x) = \int_a^b f(y | x) dy \right]$$

$$\text{we have } f(y | x) = \frac{3y^2}{1-x^3} \quad 0 < x < y < 1$$

$$\begin{aligned} \therefore f(y | x = 0.25) &= \frac{3y^2}{1-(0.25)^3} \quad 0 < y < 1 \\ &= \frac{3y^2}{1-0.016} \end{aligned}$$

$$\therefore P(Y > \frac{1}{2} | X = 0.25) = \int_{\frac{1}{2}}^1 \frac{3y^2}{1-0.016} dy = 0.89 //$$

Pb Given that the joint density function of continuous random variables X, Y is

$$f(x, y) = \begin{cases} xy^2 + \frac{x^2}{8} & , 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

find

(i) $P(X > 1)$, $P(Y < 1/2)$, $P(X > 1 | Y < 1/2)$

and $P(Y < 1/2 | X > 1)$

(2) $P(X < Y)$, $P(X + Y \leq 1)$

Soln:-

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y) dy = \int_0^1 \left(xy^2 + \frac{x^2}{8} \right) dy = \left(\frac{xy^3}{3} + \frac{x^2 y}{8} \right) \Big|_0^1$$

$$f_X(x) = \frac{x}{3} + \frac{x^2}{8}, \quad 0 \leq x \leq 2$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y) dx = \int_0^2 \left(xy^2 + \frac{x^2}{8} \right) dx = \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_0^2$$

$$f_Y(y) = 2y^2 + \frac{1}{3}, \quad 0 \leq y \leq 1$$

$$P(X > 1) = \int_1^2 f_X(x) dx = \int_1^2 \left(\frac{x}{3} + \frac{x^2}{8} \right) dx = \left[\frac{x^2}{6} + \frac{x^3}{24} \right]_1^2$$

$$P(Y < 1/2) = \int_0^{1/2} f_Y(y) dy = \int_0^{1/2} (2y^2 + 1/3) dy = \left(\frac{2y^3}{3} + \frac{y}{3} \right) \Big|_0^{1/2} = \frac{19}{24}$$

$$= \frac{1}{4}$$

$$P(X > 1 | Y < 1/2) = \frac{P(X > 1, Y < 1/2)}{P(Y < 1/2)}$$

$$P(X > 1, Y < 1/2) = \int_1^2 \int_0^{1/2} xy^2 + \frac{x^2}{8} dy dx$$

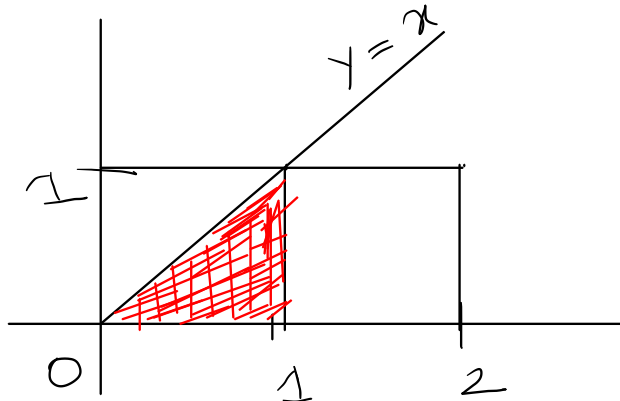
$$= \int_1^2 \left[\frac{xy^3}{3} + \frac{x^2 y}{8} \right]_0^{1/2} dx = \int_1^2 \left[\frac{x}{24} + \frac{x^2}{16} \right] dx$$

$$= 5/24$$

$$P(X > 1 | Y < 1/2) = \frac{5/24}{1/4} = 5/6$$

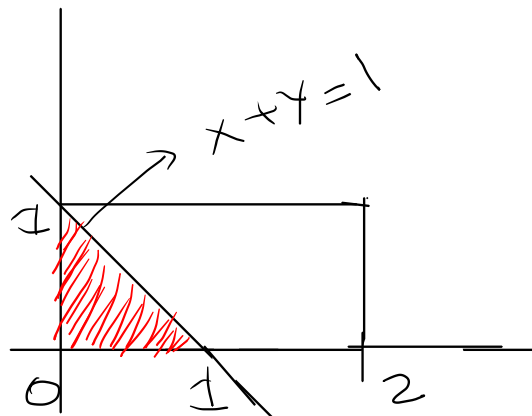
$$P(Y < 1/2 | X > 1) = \frac{P(Y < 1/2, X > 1)}{P(X > 1)} = \frac{5/24}{19/24} = 5/19$$

We need to calculate $P(X < Y)$



$$\begin{aligned} P(X < Y) &= \int_0^1 \int_0^y f(x, y) dx dy = \int_0^1 \int_0^y \left(xy^2 + \frac{x^2}{8} \right) dx dy \\ &= \int_0^1 \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_0^y dy = \int_0^1 \left[\frac{y^4}{2} + \frac{y^3}{24} \right] dy \\ &= \frac{53}{480} \end{aligned}$$

$$P(X+Y \leq 1)$$



$$\begin{aligned}
 P(X+Y \leq 1) &= \int_0^1 \int_0^{1-y} f(x,y) \, dx \, dy = \int_0^1 \int_0^{1-y} \left(xy^2 + \frac{x^2}{8} \right) \, dx \, dy \\
 &= \int_0^1 \left[\frac{x^2 y^2}{2} + \frac{x^3}{24} \right]_0^{1-y} \, dy = \int_0^1 \left[\frac{y^2 (1-y)^2}{2} + \frac{(1-y)^3}{24} \right] \, dy \\
 &= \frac{13}{480}
 \end{aligned}$$

Statistically independent

Let X and Y be two random variables with probability mass function $p(x, y)$ (if continuous then with probability density function $f(x, y)$). We say X and Y are statistically independent if

$$p(x, y) = p_X(x) \cdot p_Y(y)$$

for continuous case

$$f(x, y) = f_X(x) \cdot f_Y(y)$$

Example Consider the joint density function

$$f(x, y) = \begin{cases} \frac{x^3 y^3}{16}, & 0 \leq x \leq 2, 0 \leq y \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^2 \frac{x^3 y^3}{16} dy = \left[\frac{x^3 y^4}{64} \right]_0^2$$

$$= \frac{x^3}{4} \quad 0 \leq x \leq 2$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^2 \frac{x^3 y^3}{16} dx = \frac{y^3}{4} \quad 0 \leq y \leq 2$$

clearly

$$f(x, y) = f_X(x) \cdot f_Y(y)$$

X and Y are independent.